Diabetic Retinopathy AI Analysis Report

AI-Assisted Retinal Screening · RetinaIQ CNN + YOLO26 Pipeline

Examination Details

REPORT ID
4C86AB07

REPORT DATE
19 Oct 2026

PATIENT NAME
Surada Lohit

AGE
1

SEX
Male

DATE OF BIRTH (DOB)
2025-01-09

EMAIL
surada@gmail.com

CONTACT NO.
8079856430

ADDRESS
215, Dammala street, near water tank, kinnera theatre backside

BLOOD GROUP
A+

DR Classification

The following classification was generated by the RetinaIQ CNN severity model. This prediction represents an AI screening output and requires clinical interpretation.

Predicted Classification	Mild diabetic retinopathy
Severity Stage	Mild
Model Confidence	86.5%
Clinical Note	Early changes that should be monitored regularly.

Original Fundus Image

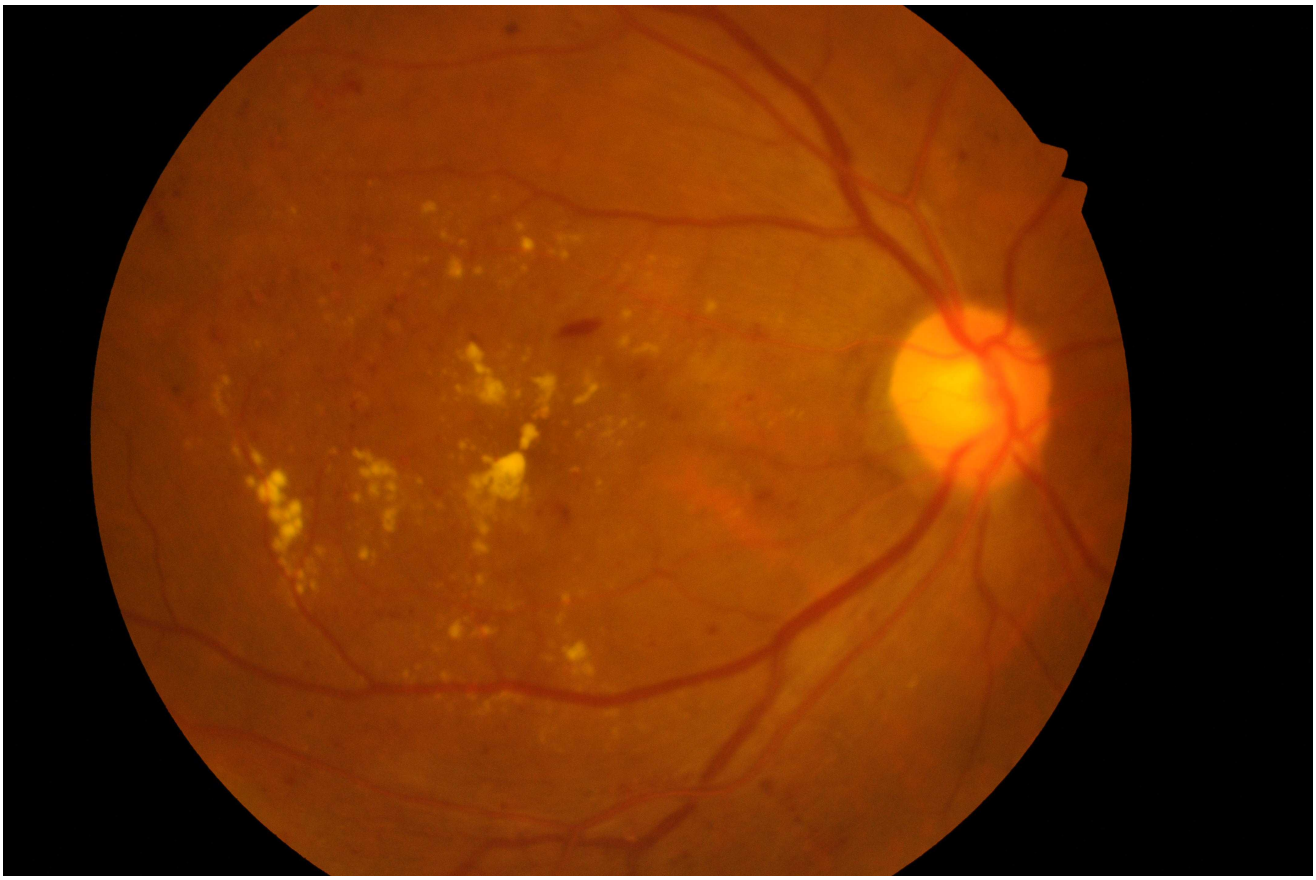


Fig 1 — Original digital retinal fundus photograph submitted for AI-assisted analysis.

Lesion Detection Results

Detected by the YOLO26 deep object-detection architecture. Counts indicate localized bounding-box detections across the retinal image field.

Lesion Type	Count	Clinical Significance
Microaneurysm	4	Focal outpouchings of capillary walls — earliest clinical sign of background DR.
Haemorrhage	10	Intraretinal bleeding indicating progressive capillary wall compromise.
Hard Exudate	109	Lipid and lipoprotein deposits resulting from chronic capillary hyperpermeability.
Soft Exudate	0	Cotton-wool spots reflecting focal ischaemic infarcts of retinal nerve fibres.
Total Detections	**123**	

Annotated Lesion Detection Map

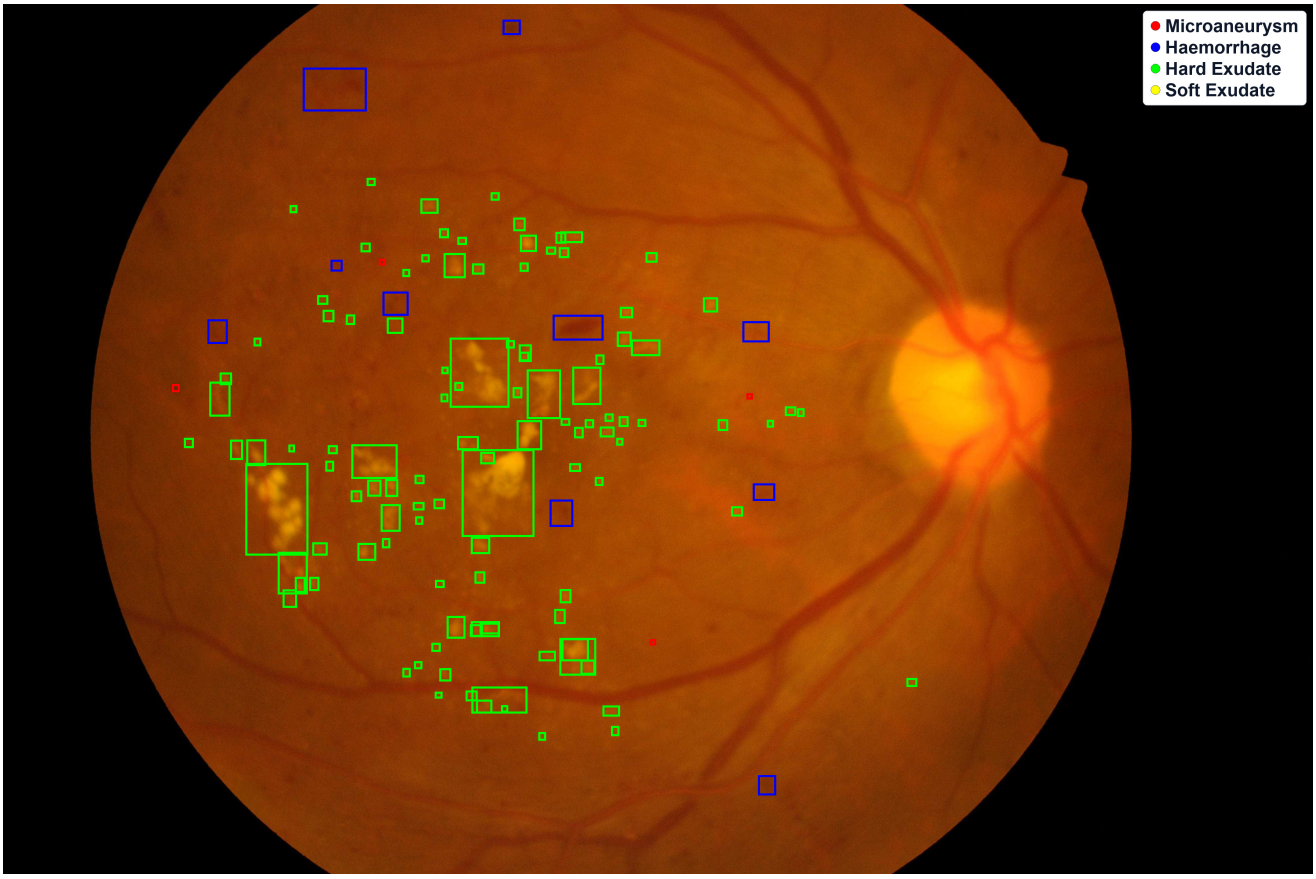


Fig 2 — YOLO26-annotated fundus image with bounding boxes identifying detected lesion regions.

Grad-CAM & Model Explainability

The visualizations below illustrate the internal spatial representations of the YOLO26 object-detection model. Each row displays the analytical image on the left with detailed clinical and model interpretation points on the right.

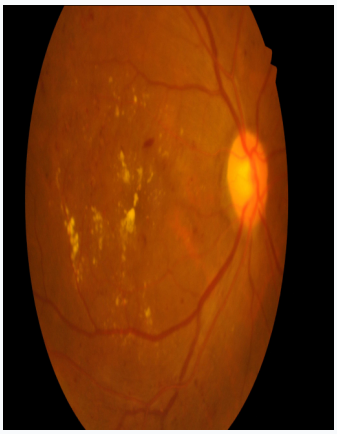


Fig 3a

Clinical Information & Model Insights:

- **Primary Role:** Unaltered color digital retinal fundus photograph submitted for diagnostic analysis.
- **Clinical Purpose:** Provides the anatomical reference standard, depicting posterior pole vascular networks, optic disc, and foveal center.
- **Diagnostic Value:** Ground truth baseline against which AI-generated heatmap activations and lesion detections are clinically verified.
- **Technical Spec:** RGB Digital Fundus Capture · Standardized Preprocessing Entry Pipeline.

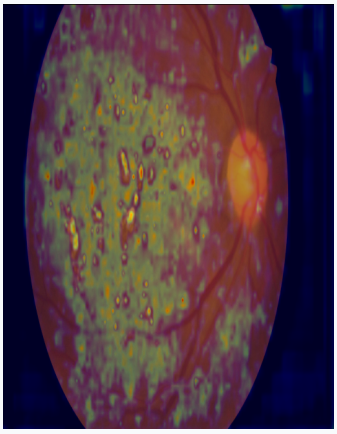


Fig 3b

Clinical Information & Model Insights:

- **Primary Role:** Gradient-weighted class activation map highlighting retinal regions exerting the highest neural weight.
- **Clinical Purpose:** Backpropagates gradients from the detection head to layer-16 feature maps to reveal decision-critical lesion clusters.
- **Key Observations:** Warm spectrum (red/yellow) marks focal pathological lesion clusters; cool colors denote background parenchymal regions.
- **Scope Clarification:** Specific to YOLO26 lesion detection; does not independently explain the separate CNN severity classification.

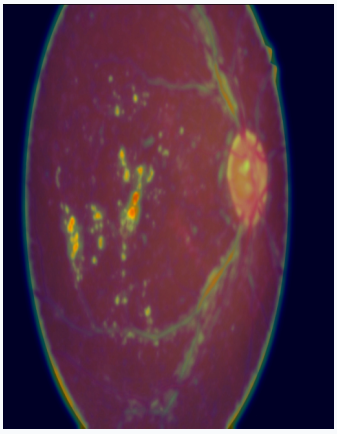


Fig 3c

Clinical Information & Model Insights:

- **Primary Role:** Adaptive histogram equalization (CLAHE) local texture and contrast differential map.
- **Clinical Purpose:** Amplifies microvascular contours to differentiate faint microaneurysms and haemorrhages from uneven retinal illumination.
- **Key Observations:** Structural contrast enhancement (not Grad-CAM). Differentiates sharp lipid borders of hard exudates from cotton-wool spots.
- **Technical Spec:** CLAHE Contrast Differential Filtering · Adaptive Local Histogram Clip.

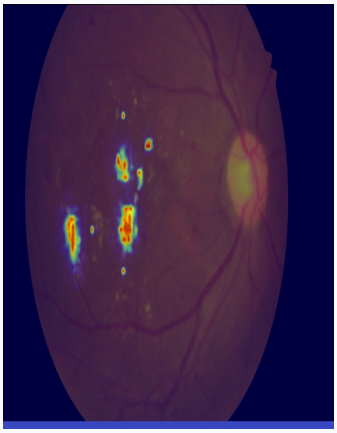


Fig 3d

Clinical Information & Model Insights:

- **Primary Role:** Direct visual representation of deep intermediate activations extracted from the detection backbone.
- **Clinical Purpose:** Demonstrates neural transformation of raw pixel intensities into composite spatial lesion features.
- **Key Observations:** Validates that spatial coordinates of retinal abnormalities remain preserved before bounding-box regression heads.
- **Technical Spec:** Intermediate Backbone Activations · Multi-Channel Spatial Tensor Projection.

Recovery & Prevention Priorities

The following priorities provide educational guidance based on the AI screening result. This section does not constitute a clinical prescription or personalized treatment plan.

1.	Schedule a comprehensive dilated eye exam every 6–12 months to monitor progression of mild diabetic retinopathy.
2.	Maintain tight glycemic control—aim for an A1C below 7% if possible—to slow retinal changes.
3.	Adopt a heart■healthy lifestyle: balanced diet, regular exercise, and smoking cessation to support overall vascular health.

Seek Urgent Eye Care Immediately: If you experience sudden vision loss, a sudden shower of new floaters, flashes of light, a dark shadow or curtain across your field of vision, or severe eye pain, seek emergency ophthalmology care immediately.

Medicines & Eye-Care Guidance

The items listed below are educational references for discussion with your ophthalmologist — not prescriptions or treatment regimens. All therapeutic decisions must be determined by a licensed ophthalmologist following a comprehensive in-person clinical evaluation.

Metformin – commonly used to improve insulin sensitivity and help lower blood glucose levels.

Clinician discussion reference

ACE inhibitor (e.g., lisinopril) – often prescribed to manage blood pressure and protect kidney and retinal blood vessels.

Clinician discussion reference

Statin (e.g., atorvastatin) – used

Clinician discussion reference